

INTERNSHIP REPORT

On

JAVA FULL STACK DEVELOPMENT

Submitted

To

**School of Computer Science and Engineering
IILM University, Greater Noida, U.P.**

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

Roll Number of Student: 2410030430

Name of Student: NIDHI KUMARI

Batch: 2024-2028

CANDIDATE'S DECLARATION

I, Nidhi Kumari, do hereby declare that the internship report titled "Java Full Stack Development" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature: Nidhi Kumari

Student Name: Nidhi Kumari

Roll No.: 2410030430

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank EduSkills Academy and AICTE for providing the opportunity to undertake the AICTE-EduSkills Virtual Internship Program in Java Full Stack Development (January 2026 - March 2026), and my faculty at IILM University, Greater Noida, for their continuous encouragement and support throughout the duration of this internship.

I express my gratitude to my parents for their blessings.

Signature: Nidhi Kumari

Student Name: Nidhi Kumari

Roll No.: 2410030430

TABLE OF CONTENTS

Chapter No.	Chapter Name	Page No.
	Cover Page	
1.	Candidate's Declaration	2
2.	Acknowledgment	3
3.	Internship Offer Letter	
4.	Internship Completion Certificate	
5.	Project Description	8
	5.1 Introduction	8
	5.2 Organization Profile	8
	5.3 Problem Statement	9
	5.4 Project Objectives	9
	5.5 Scope of the Project	9
	5.6 Technologies and Tools Used	10
	5.7 System Architecture / Learning Architecture	10
	5.8 Methodology	11
	5.9 Expected Outcomes	12
	5.10 Certificate of Completion	12
6.	Bibliography / References	13

3. INTERNSHIP OFFER LETTER

Nidhi Kumari was selected for the 10-Week AICTE - EduSkills Virtual Internship Program in the domain of Java Full Stack Developer, for the duration January 2026 to March 2026 (Ref No.: C15Y2026M081355845, AICTE Student ID: STU6832c2f2e957c1748157170). The offer letter issued by EduSkills Academy is reproduced below.



Ref No: C15Y2026M081355845

Internship Offer Letter

Student Name : Nidhi Kumari
Stream : B.Tech
Branch : B.Tech(Computer Science and Engineering)
Institute Name : IILM University, Greater Noida
Internship Domain : Java Full Stack Developer
Internship Duration : January 2026 - March 2026
AICTE Student ID : STU6832c2f2e957c1748157170

We are pleased to inform you that you have been selected for the 10-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

The week wise Structure of the Internship is as Follows:

Week	Module	Description
Week 1	HTML	Learn HTML fundamentals including page structure, semantic elements, forms, tables, and building the foundation of web pages.
Week 2	CSS	Master CSS styling concepts including selectors, box model, positioning, flexbox, grid, and responsive web design.
Week 3	Bootstrap	Build responsive and mobile-first interfaces using Bootstrap components, grid system, and utility classes.
Week 4	JavaScript & jQuery	Learn JavaScript fundamentals, DOM manipulation, events, and enhance development efficiency using jQuery for interactive web applications.
Week 5	Core Java	Understand Java fundamentals including OOP concepts, data types, collections, exception handling, and file handling.
Week 6	Advanced Java	Explore advanced Java concepts including JDBC, Servlets, multithreading, and enterprise-level application development principles.
Week 7	Spring Framework & Spring Boot	Develop enterprise applications using Spring Framework and Spring Boot, covering dependency injection, REST APIs, and application configuration.
Week 8	Hibernate & MySQL	Learn ORM concepts using Hibernate and integrate MySQL database for data persistence and CRUD operations.
Week 9	Git & Version Control	Manage source code using Git and GitHub, implement branching strategies
Week 10	Assessment	Assessment

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.



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4. INTERNSHIP COMPLETION CERTIFICATE

**CERTIFICATE
OF INTERNSHIP**





PRESENTED TO

Nidhi Kumari

has successfully achieved the Internship Credential by completing a 8-week program in the domain of

Java Full Stack

During the internship, the learner demonstrated proficiency in:

- Introduction to HTML and Web Page Structure
- CSS Fundamentals for Web Design & Bootstrap: Responsive Web Design Framework
- JavaScript: Interactive Web Development & jQuery: Simplifying JavaScript Development
- Core Java: Fundamentals of Java Programming & Advanced Java: Enterprise Application Development
- Spring Framework: Enterprise Java Development
- Spring Boot: Simplifying Java Application Development
- Hibernate: Object-Relational Mapping in Java
- MySQL & Git & Version Control

This certification recognizes the successful completion of the internship requirements.



Issued on: Mar 20, 2026
Certificate ID: 2026-E2CF1F0C43



Milu Swain
General Manager
(Learning & Development)
EduSkills

5. PROJECT DESCRIPTION

5.1 Introduction

This report presents an account of the 10-Week AICTE - EduSkills Virtual Internship Program in Java Full Stack Development, undertaken from January 2026 to March 2026 under the domain of Java Full Stack Developer (Ref No.: C15Y2026M081355845, AICTE Student ID: STU6832c2f2e957c1748157170). The internship was designed as a structured, module-based virtual learning program covering the complete web and enterprise application development stack, starting from front-end fundamentals such as HTML, CSS and JavaScript, and progressing through Core and Advanced Java, the Spring ecosystem (Spring Framework, Spring Boot), Hibernate for object-relational mapping, and MySQL for relational database management, along with Git and GitHub for version control, culminating in a final assessment test in Week 10. The program was undertaken to strengthen practical, industry-relevant full stack development skills alongside ongoing academic coursework in the B.Tech Computer Science and Engineering program at IILM University, Greater Noida.

As a full stack development enthusiast with prior hands-on experience in building projects using the MERN/Next.js stack, this internship provided an opportunity to gain equivalent depth in the Java enterprise ecosystem, broadening exposure across multiple full stack technology stacks used in the software industry. The internship was unpaid (no stipend) and was structured entirely around weekly assessments, project documentation submissions for select modules, and a final assessment test, as outlined in the offer letter issued by EduSkills Academy.

5.2 Organization Profile

EduSkills Academy is an ISO 9001:2015 and ISO 20000-1:2018 certified ed-tech training organization, recognized as a Great Place to Work, that offers industry-aligned skill development and internship credential programs across domains such as full stack development, cloud computing, data science, and emerging technologies. This particular internship was delivered under the AICTE (All India Council for Technical Education) - EduSkills Virtual Internship Program, offered under the patronage of AICTE's National Internship Portal, with the program supported by EduSkills Academy and led by Mr. Bibek Ranjan, Director, EduSkills. The organization partners with academic institutions to deliver structured, certification-backed internship programs

that combine self-paced learning modules with weekly skill assessments, enabling students to gain verifiable, hands-on proficiency in current industry tools and frameworks.

5.3 Problem Statement

Modern enterprise software development relies heavily on Java-based full stack frameworks for building secure, scalable, and maintainable web applications. While front-end technologies and basic scripting are increasingly accessible through self-learning, proficiency in enterprise-grade back-end frameworks such as Spring, Spring Boot, and Hibernate, combined with structured database design, requires guided and progressive learning. The problem addressed by this internship was to bridge this specific skill gap by following a structured, week-wise curriculum that builds full stack competency layer by layer, from web fundamentals to enterprise Java application development, validated through weekly assessments and a final assessment test.

5.4 Project Objectives

- To understand and apply core front-end web development concepts using HTML, CSS, Bootstrap, JavaScript, and jQuery.
- To build a strong foundation in Core Java and Advanced Java programming concepts.
- To learn enterprise application development using the Spring Framework and Spring Boot.
- To understand Object-Relational Mapping (ORM) concepts and implement them using Hibernate.
- To design and manage relational databases using MySQL.
- To gain practical experience with Git and GitHub version control workflows, including branching strategies used in collaborative software development.
- To integrate front-end, back-end, and database layers into a cohesive full stack development skill set.

5.5 Scope of the Project

The scope of this internship was limited to structured, module-based skill acquisition across the Java full stack ecosystem over 10 weeks rather than the delivery of a single

standalone application. It covered the front-end layer (HTML, CSS, Bootstrap, JavaScript, jQuery), the back-end layer (Core Java, Advanced Java, Spring Framework, Spring Boot), the persistence layer (Hibernate, MySQL), and supporting development practices (Git and GitHub version control), concluding with a final assessment test in Week 10. The concepts and frameworks studied are directly applicable to building production-grade, enterprise Java web applications, and form a foundation that can be extended into independent project work in the future.

5.6 Technologies and Tools Used

Table 4.1: Technologies and tools covered during the internship

Layer	Technologies / Tools
Front-End	HTML, CSS, Bootstrap, JavaScript, jQuery
Core Programming	Core Java, Advanced Java
Back-End Frameworks	Spring Framework, Spring Boot
ORM (Object-Relational Mapping)	Hibernate
Database	MySQL
Version Control	Git

5.7 System Architecture / Learning Architecture

The internship curriculum followed a layered full stack architecture consistent with standard enterprise Java applications:

- Presentation Layer: Built using HTML, CSS, Bootstrap for responsive layouts, and JavaScript/jQuery for client-side interactivity.
- Application/Business Layer: Implemented using Core Java and Advanced Java concepts, structured through the Spring Framework and Spring Boot for dependency injection, REST API development, and application configuration.
- Persistence Layer: Managed through Hibernate ORM, mapping Java objects to relational database tables.
- Database Layer: MySQL used as the relational database management system for data storage and retrieval.
- Version Control Layer: Git used throughout for tracking changes and managing code across modules.

This layered approach mirrors the standard three-tier architecture (presentation, business logic, and data) widely used in real-world enterprise Java applications.

5.8 Methodology

The internship followed the week-wise structured learning plan prescribed by EduSkills Academy, with each week focused on a specific module and reinforced through weekly assessments:

- Week 1 - HTML: Page structure, semantic elements, forms, and tables, forming the foundation of web pages.
- Week 2 - CSS: Selectors, the box model, positioning, Flexbox, Grid, and responsive web design.
- Week 3 - Bootstrap: Responsive and mobile-first interfaces using Bootstrap components, the grid system, and utility classes.
- Week 4 - JavaScript & jQuery: JavaScript fundamentals, DOM manipulation, events, and jQuery for interactive web applications.
- Week 5 - Core Java: Java fundamentals including OOP concepts, data types, collections, exception handling, and file handling.
- Week 6 - Advanced Java: JDBC, Servlets, multithreading, and enterprise-level application development principles.
- Week 7 - Spring Framework & Spring Boot: Dependency injection, REST APIs, and application configuration for enterprise applications.
- Week 8 - Hibernate & MySQL: ORM concepts using Hibernate, integrated with a MySQL database for data persistence and CRUD operations.
- Week 9 - Git & Version Control: Managing source code using Git and GitHub, and implementing branching strategies.
- Week 10 - Final Assessment: A comprehensive assessment test covering all modules, in addition to the weekly assessments and project documentation submitted through the internship.

Each module combined conceptual learning with hands-on practice exercises, reinforcing theoretical understanding through applied coding tasks, weekly assessments, and periodic project documentation submissions.

5.9 Expected Outcomes

- A working understanding of responsive front-end development using HTML, CSS, Bootstrap, JavaScript, and jQuery.
- Proficiency in Core and Advanced Java programming concepts applicable to enterprise application development.
- Practical familiarity with building applications using the Spring Framework and Spring Boot.
- Understanding of ORM concepts and hands-on exposure to Hibernate.
- Ability to design and query relational databases using MySQL.
- Working knowledge of Git and GitHub for version control and branching strategies in a development workflow.
- A broadened full stack skill set spanning both the Java enterprise ecosystem and the JavaScript/Next.js ecosystem used in prior independent projects.

5.10 Certificate of Completion

The Internship Credential Certificate issued by EduSkills Academy on successful completion of the AICTE - EduSkills Virtual Internship Program in Java Full Stack is attached in Chapter 4 of this report (Certificate ID: 2026-E2CF1F0C43, issued on March 20, 2026). The corresponding internship offer letter (Ref No.: C15Y2026M081355845) is attached in Chapter 3.

6. BIBLIOGRAPHY / REFERENCES

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